# 3-Variable & 4-Variable Karnaugh Maps

## 1. The 3-Variable K-Map

### Theory:

A 3-variable K-Map is drawn for a Boolean expression with three variables (e.g., $A, B, C$). It contains $2^3 = 8$ cells, arranged in a $2 \times 4$ grid.

Notice the Gray Code sequence ( `00, 01, 11, 10` ) along the top axis. This creates a specific numbering layout for the minterms.

### 3-Variable Grid Layout:

| $A \backslash BC$ | $00(\bar{B}\bar{C})$ | $01(\bar{B}C)$ | $11(BC)$ | $10(B\bar{C})$ |
|---|---|---|---|---|
| $0(\bar{A})$ | $m_0$ | $m_1$ | $m_3$ | $m_2$ |
| $1(A)$ | $m_4$ | $m_5$ | $m_7$ | $m_6$ |

*(Crucial Note: Remember that the columns jump from 1 to 3, then back to 2. The same happens on the bottom row from 5 to 7, then 6!)*

## Example 1: 3-Variable SOP

Simplify $F(A, B, C) = \Sigma m(0, 1, 6, 7)$

### The Map:

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | **1** | **1** | 0 | 0 |
| 1 | 0 | 0 | **1** | **1** |

- **Group 1 (Top Left):** Covers $m_0$ and $m_1$. $C$ changes (0 to 1), so it drops. Result: $\bar{A}\bar{B}$.
- **Group 2 (Bottom Right):** Covers $m_7$ and $m_6$. $C$ changes (1 to 0), so it drops. Result: $AB$ .
- **Final Expression:** $F = \bar{A}\bar{B} + AB$

## Example 2: 3-Variable POS (Grouping 0s)

Simplify $f(A, B, C) = \Pi M(1, 2, 3, 6, 7)$

### The Map:

| $A\backslash BC$ | $00(B+C)$ | $01(B+\bar{C})$ | $11(\bar{B}+\bar{C})$ | $10(\bar{B}+C)$ |
|---|---|---|---|---|
| $0(A)$ | 1 | **0** | **0** | **0** |
| $1(\bar{A})$ | 1 | 1 | **0** | **0** |

- **Group 1 (Square of four):** Covers $M_3, M_2, M_7, M_6$. $A$ changes, $C$ changes. Only $\bar{B}$ stays constant. Result: $\bar{B}$.
- **Group 2 (Pair):** Covers $M_1, M_3$. $C$ changes. Result: $(A+\bar{C})$.
- **Final Expression:** $f = \bar{B} \cdot (A+\bar{C})$

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## 2. The 4-Variable K-Map

### Theory:

A 4-variable K-Map is used for expressions with four variables (e.g., $A, B, C, D$). It contains $2^4 = 16$ cells, arranged in a $4 \times 4$ grid. Gray code is applied to both the vertical and horizontal axes.

### 4-Variable Grid Layout:

| $AB\backslash CD$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | $m_0$ | $m_1$ | $m_3$ | $m_2$ |
| 01 | $m_4$ | $m_5$ | $m_7$ | $m_6$ |
| 11 | $m_{12}$ | $m_{13}$ | $m_{15}$ | $m_{14}$ |
| 10 | $m_8$ | $m_9$ | $m_{11}$ | $m_{10}$ |

## Example: 4-Variable SOP with Don't Cares

Simplify $F(A,B,C,D) = \Sigma m(0,2,8,10,14) + \Sigma d(5,15)$

### The Map:

| $AB\backslash CD$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | **1** | 0 | 0 | **1** |
| 01 | 0 | **X** | 0 | 0 |
| 11 | 0 | 0 | **X** | **1** |
| 10 | **1** | 0 | 0 | **1** |

### Theory - Map Rolling (Wrap-around):

In a 4-variable map, the top row is adjacent to the bottom row, and the left column is adjacent to the right column. You can group "through the walls" of the map.

- **Group 1 (The Four Corners):** $m_0, m_2, m_8, m_{10}$ wrap around all four edges to form a single group of 4. $A$ changes (0 to 1), $C$ changes (0 to 1). We are left with $\bar{B}\bar{D}$.
- **Group 2 (Vertical Pair):** $m_{14}$ and the "Don't Care" $m_{10}$ are grouped. $C$ changes. We are left with $AC\bar{D}$. *(Note: We ignore the Don't Care $X_5$ because it doesn't help us make a bigger group).*
- **Final Expression:** $F = \bar{B}\bar{D} + AC\bar{D}$